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E-waste recycled materials as efficient catalysts for renewable energy technologies and better environmental sustainability

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Abstract

Waste from electrical and electronic equipment exponentially increased due to the innovation and the ever-increasing demand for electronic products in our life. The quantities of electronic waste (e-waste) produced are expected to reach 44.4 million metric tons over the next five years. Consequently, the global market for electronics recycling is expected to reach \$65.8 billion by 2026. However, electronic waste management in developing countries is not appropriately handled, as only 17.4% has been collected and recycled. The inadequate electronic waste treatment causes significant environmental and health issues and a systematic depletion of natural resources in secondary material recycling and extracting valuable materials. Electronic waste contains numerous valuable materials that can be recovered and reused to create renewable energy technologies to overcome the shortage of raw materials and the adverse effects of using non-renewable energy resources. Several approaches were devoted to mitigate the impact of climate change. The cooperate social responsibilities supported integrating informal collection and recycling agencies into a well-structured management program. Moreover, the emission reductions resulting from recycling and proper management systems significantly impact climate change solutions. This emission reduction will create a channel in carbon market mechanisms by trading the CO₂ emission reductions. This review provides an up-to-date overview and discussion of the different categories of electronic waste, the recycling methods, and the use of high recycled value-added (HAV) materials from various e-waste components in green renewable energy technologies.

Keywords: E-waste; Energy conversion; Energy storage; Global market; Green hydrogen; Hydrometallurgy; Metal recovery; Recycling; Supercapacitor; Sustainability.

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Figures

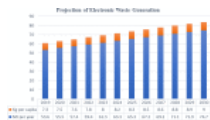


Fig. 1 Projection of e-waste generation (Forti...



Fig. 2 Leading countries in the generation...



Fig. 3 WEEE categories according to the...



Fig. 4 Closed Loop supply chain of...



Fig. 5 Design application for the collection...



Fig. 6 Typical direct, pyrometallurgical, and hydrometallurgical...

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